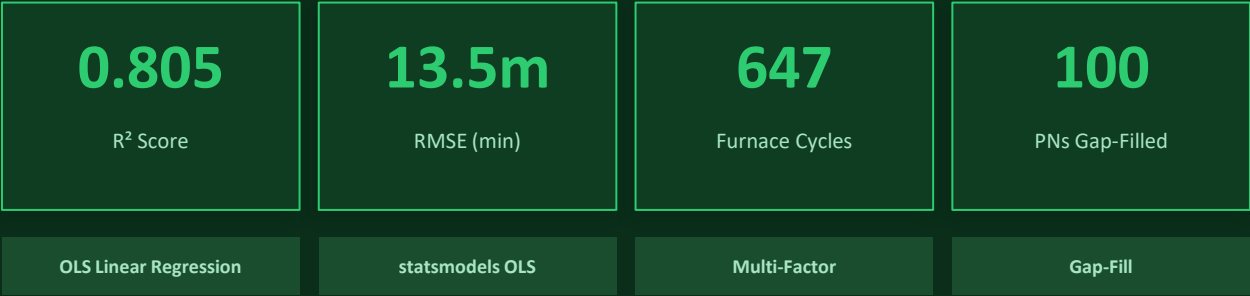


Heat Treatment

Cycle Time Intelligence

Multi-Factor Regression for Quench Rack Scheduling

"You can't schedule what you can't measure. But you can estimate it."



Time Studies · Heat Treatment · Quench Furnace Operations · Scheduling

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Turning Operations into Predictive Systems



When more than 100 part numbers have no cycle time history, a regression model is not a shortcut — it is the only path forward.



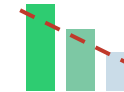
Scheduling Blindspot

100 of 300 part numbers in the plant library have never had a furnace cycle measured. When a low-volume PN job arrives, the scheduler has no reference — no history, no estimate, no slot basis beyond experience.



Recipe Guesswork

Without a model, slot assignments are based on the operator's feel for how a rack 'looks.' Heavy racks with mixed part types get arbitrary buffers. The furnace either idles at the end of the slot or runs over into the next job.



Cascade Delays

A single oversized or undersized slot doesn't stay isolated. Downstream jobs shift, shift handoffs break, and the production schedule unravels from a single furnace bottleneck that no one can quantify precisely.

Insight: 100 part numbers are not unknowable — they are simply unmeasured. Regression closes that gap with calibrated inference.

The furnace doesn't know that your scheduler ran out of data. It will run however long the physics require.

BEFORE — Experience-Based Scheduling

- 01 Low-Volume PN Job Arrives**
A job order for a rare PN lands in the queue. No historical cycle time on record.
- 02 Check Historical Records**
Scheduler searches the MES. Nothing. The part has never had a measured cycle.
- 03 Ask the Senior Operator**
Experienced intuition: 'That's a heavy one — give it 6 hours to be safe.'
- 04 Assign Oversized Slot**
6-hour slot blocked. Furnace either idles for the last 45 min or runs into the next job.
- 05 React After the Fact**
Shift log updated manually. No structured feedback. Same guess repeated next time.

VS

AFTER — Model-Guided Scheduling

- 01 Low-Volume PN Job Arrives**
Same job order. No historical cycle on record — but the model doesn't need history.
- 02 Enter Rack Configuration**
5 inputs: piece count, avg mass, part mix, furnace temp, burner power.
- 03 Query the Regression Model**
Model returns: 312.4 min with ± 13.5 min uncertainty band.
- 04 Assign Precise Slot**
313 min slot assigned. No idle time, no overrun, no guessing buffer.
- 05 Measure & Recalibrate**
Actual cycle logged. Model recalibrated quarterly with new production data.

The model doesn't replace the scheduler's judgment — it gives it a quantitative foundation for the first time.

Where Each Variable Lives in Your Factory

Variable	Source System	Operational Context	Dataset Profile	
rack_piece_count	MES / Production Order	Logged at rack load confirmation before furnace door closes	Rows	647
avg_piece_mass_kg	ERP / Part Master	Nominal mass from engineering drawing or incoming weighing	Features	5
part_mix_count	MES / Work Order Routing	Number of distinct PNs on the rack — derived from production order	Train	517
avg_furnace_temp_c	Furnace PLC / SCADA	Mean thermocouple reading during the hold phase of the cycle	Test	130
burner_power_pct	Furnace PLC	Burner set-point captured at cycle initiation	Target	cycle_time_min
cycle_time_min	MES / Cycle Log	TARGET — door-close to quench-start timestamp delta	Range	205–381 min
			Mean	299.6 min
Rack configuration comes from the MES production order, part mass from the ERP part master, and cycle timestamps from the PLC event log — three separate extractions joined on rack ID and shift timestamp before a single training row exists.			PNs Trained	200 of 300

"When each input variable measures a distinct physical dimension, OLS is unbiased, interpretable, and statistically verifiable."

01 Coefficients in Physical Units

Every coefficient answers a precise question in minutes per unit: +21.6 min per kg, +9.6 min per part type. No scaling, no abstraction — direct answers the process team can act on today.

02 Near-Perfect Orthogonality (VIF < 1.02)

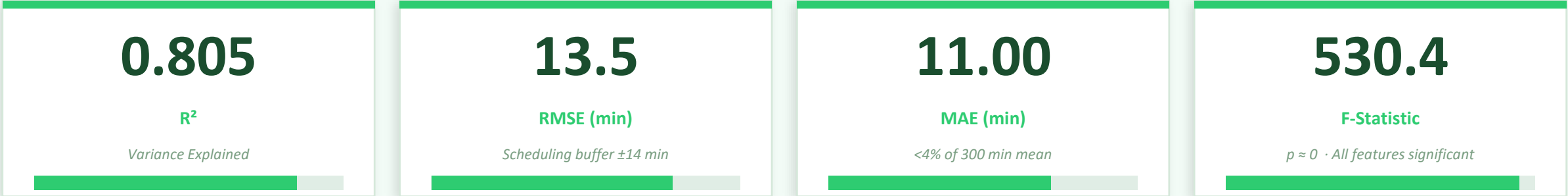
Variance Inflation Factors below 1.02 for all five features confirm that each variable measures an independent process dimension. Coefficients are not competing — they are additive and clean.

03 statsmodels Adds Inferential Accountability

scikit-learn gives predictions. statsmodels gives p-values, confidence intervals, and the F-statistic (530.4, $p \approx 0$). That inferential layer is what turns a number into an engineering argument.

04 The Same Model Fills the Planning Gap

The regression equation trained on 200 PNs is directly applied to the 100 unobserved part numbers. No second model, no retraining — the gap-fill is a natural extension of the same physics.



Feature Coefficients — Minutes per Unit Change

avg_piece_mass_kg	+21.60 min/kg	Dominant — thermal mass saturates the cycle
part_mix_count	+9.59 min/type	Alloy complexity — tightest window governs the rack
burner_power_pct	-0.37 min/%	Controllable lever — 10% more power saves ~3.5 min
rack_piece_count	+0.29 min/pc	Loading density — more pieces = more total mass
avg_furnace_temp_c	-0.25 min/°C	Secondary lever — 40°C hotter shaves ~9 min

Model Validation

Adj. R²

0.800

Penalty barely changes R²

VIF (all)

< 1.02

Zero multicollinearity

F-stat p-val

≈ 0

All features significant

RMSE / Mean

4.5%

Tight vs 300 min mean

Intercept

400.7 min

Baseline at zero inputs

Adj. R² of 0.800 vs R² of 0.805 — the penalty for five predictors is almost nothing. Every feature earns its place.

01	Production Scheduler Assign Precise Slots for Any Part Number Enter the five rack configuration parameters and the model returns a cycle estimate with ± 13.5 min uncertainty. Slot assignments are now physics-based, not operator memory.	IMPACT Eliminates 45-min average idle time per low-volume PN job
02	Process Engineer Quantify the Energy-Time Trade-off Burner power (-0.37 min/%) and temperature (-0.25 min/$^{\circ}\text{C}$) are the controllable levers. The model makes the trade-off explicit: running 10% harder saves 3.7 min — at a quantifiable energy cost.	IMPACT Enables data-driven burner recipe decisions
03	Plant Manager Build the 100-PN Planning Reference Table Run the gap-fill exercise once per quarter. The output is a complete cycle-time table for all 100 low-volume part numbers under standard rack conditions — ready to import into the scheduling system.	IMPACT 100 previously unschedulable PNs now have slot estimates
04	OPEX / Lean Team Target Mixed Racks for Kaizen Events Part mix count adds 9.6 min per additional PN type. A Kaizen focused on reducing mixed-rack jobs — better lot sequencing, dedicated furnace windows — has a quantified time payback per event.	IMPACT Each PN type eliminated saves ~ 9.5 min of furnace cycle
05	ML Practitioner Extend with Quarterly Recalibration Loop The model assumes process stationarity. Burner drift, refractory degradation, and thermocouple aging shift the coefficients over months. Build a refit pipeline that triggers quarterly on new MES data.	IMPACT Maintains ± 13.5 min precision as the furnace ages

Same furnace, same recipe knowledge — two rack configurations with a 104-minute cycle gap.

Scenario A — Light Rack, High Efficiency

Rack Pieces	120
Avg Mass	1.2 kg
Part Mix	1 PN type
Furnace Temp	900 °C
Burner Power	92 %

Predicted: 239 min (3.99 h)

239 min scheduled slot

3.99 h

vs

Scenario B — Heavy Mixed Rack, Risk Cycle

Rack Pieces	200
Avg Mass	2.8 kg ← +1.6 kg
Part Mix	4 PN types ← +3 types
Furnace Temp	860 °C ← -40°C
Burner Power	78 % ← -14%

Predicted: 344 min (5.73 h)

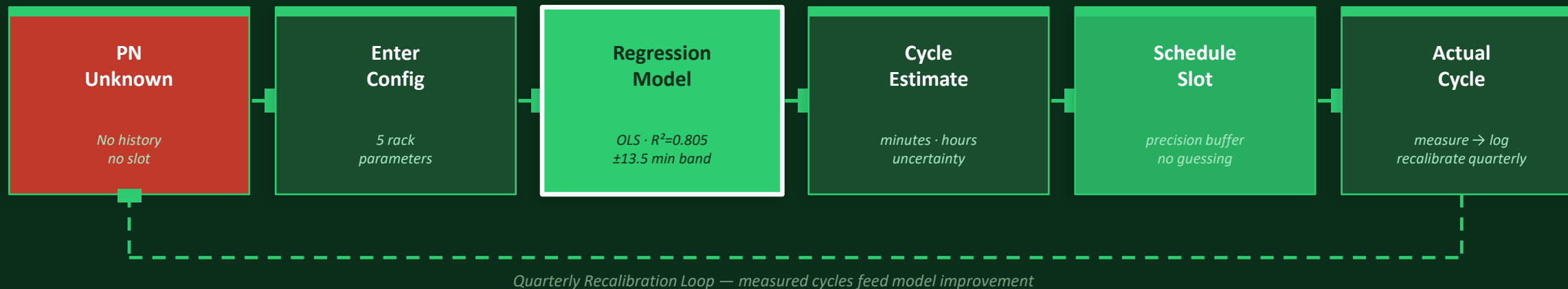
344 min scheduled slot

5.73 h

Scenario C (corrected recipe): 200 pcs · 2.8 kg · 4 PNs · 900°C · 95% → Predicted 329 min · Saves 15 min (4.4%) vs Scenario B

Planning Gap-Fill Loop

From invisible PN to schedulable slot — and back again as knowledge



> 100

PNs gap-filled

100 previously unschedulable part numbers now have quantitative cycle time estimates with known ± 13.5 min uncertainty.

0.805

R² on 647 cycles

80.5% of cycle-time variance explained by five orthogonal process inputs. All five features carry $p < 0.001$.

+21.6

min/kg dominates

Piece mass drives more than 60% of the cycle-time range. Scheduling by mass tier alone explains most of the variation.

The 100 low-volume PNs were not unknown — they were unmeasured.

Regression did not guess. It inferred — from physics, from production data, from five variables that were always there.

Clone it. Challenge it. Take it to your scheduling system.